

Chapter 75 — Surgery for Erectile Dysfunction

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- **Modern penile implants were first marketed for erectile dysfunction (ED) over 50 years ago:** the **three-piece inflatable implant of Scott (1972)** and the **semirigid rod prosthesis of Small and Carrion (1978)**. Before them, **rudimentary penile stiffeners** gave variable, inconsistently reliable support.
- At introduction they were **the only effective treatment for ED**; later came **oral medication, intracorporal injections, intraurethral pellets and vacuum erection devices (VEDs)**, with varying success and acceptance.
- **Improvements in design, surgical placement and repair technique** have increased device longevity and functionality.
- **Patient and partner satisfaction is high (80–90%); repairs are needed in about 15% of devices at 5 years and 30% at 10 years** (long-term series).
- Despite this, **implants remain a secondary treatment of ED** — chosen by **strongly motivated men** who have **failed, are contraindicated to, or find unacceptable** the less invasive options.
- **Lue's goal-directed approach:** outline all treatments → offer a trial of each as the patient wishes → settle on the modality that is **effective and appealing**.
- **Extensive testing for the exact cause of ED is costly and does not change therapeutic outcome**; it is reserved for rare and unusual cases.

Concept — *The implant is the **most effective but last-line** ED treatment: the goal-directed model selects by patient acceptance of successive options, not by aetiological work-up.*

TYPES OF IMPLANTS AVAILABLE

Historical and Market Background

- **Early 1970s: two types** — the **Scott inflatable** and the **Small–Carrion sponge-filled semirigid rod**. Design improvements and competing vendors (at one point **five companies**) fuelled growth.
- **Sales grew almost exponentially through the 1980s and early 1990s, peaking in 1996 at about 33,000 units.**
- **1996: preliminary sildenafil citrate (Viagra) data presented** → sales fell to a **nadir of about 12,000 units the following year.**
- **Phosphodiesterase type 5 inhibitors (PDE5Is) are not universally effective** → implant sales gradually recovered.
- **Three United States vendors today:** **Boston Scientific (Marlborough, MA), Coloplast (Minneapolis, MN) and Rigicon (Ronkonkoma, NY)**; others exited through falling sales or acquisition. All three sell internationally.
- **About 60,000 implants per year worldwide, about 30,000 in the United States. South Korea: about 8000 per year — almost twice the United States rate per capita. All other countries, including China and India: fewer than 1000 per year.**
- **Larger foreign vendors:** **Zephyr Surgical Instruments (Geneva) and Promedon (Córdoba, Argentina)**; some foreign vendors do not distribute in the United States.

*The Two Categories: Hydraulic and Semirigid Rod***HYDRAULIC VERSUS SEMIRIGID ROD IMPLANTS**

Feature	Three-piece inflatable (hydraulic)	Malleable semirigid rod
Components	Two cylinders (one per corpus cavernosum) → tubing → scrotal pump → abdominal reservoir	Two solid bendable rods only
Mechanism	Pump transfers fluid between reservoir and cylinders; cylinders act like inner tubes in a bicycle tyre , filling the erectile bodies and stretching the elastic tunica albuginea	Bends like electric wire ; firmness from intrinsic rod rigidity plus dead space (less dead space → greater rigidity)
Flaccid state	Very loose — fluid totally evacuated	Permanently firm; fixed girth sizes
Mechanical failure	Commonest complication of implant surgery	Breakage exceedingly rare
Best for	Most popular device with implanters and patients	Limited mental or manual dexterity , or where repair of a hydraulic device would be undesirable

- The defining advantage of the three-piece device is the combination of exceptional rigidity with true flaccidity.
- Two-piece inflatables (pump plus two cylinders) were marketed for four decades but **never popular**, especially with experienced implanters; the last, the **Boston Scientific Ambicor**, has been discontinued.
- Semirigid rods are placed slightly shorter than the measured corporal length → easier bending and less recoil.

Three-Piece Inflatable Models

- Expansion and width by model are in Table 75.1. **AMS (American Medical Systems) 700 LGX** lengthens by up to 25%; the **CXR is the narrower CX**; the cylinders are illustrated as three 18-cm cylinders at full inflation.
- **Rigicon Infla 10**: preliminary durability, infection-reoperation and mechanical reliability **comparable to AMS 700 and Titan**; expected in the United States soon.

*Malleable Semirigid Rods***MALLEABLE SEMIRIGID ROD IMPLANTS COMPARED**

Device	Vendor	Core	Cover
Tactra	Boston Scientific	Nitinol (nickel–titanium alloy)	Silicone
Genesis	Coloplast	Braided silver wire strands	Silicone
Rigi 10	Rigicon	Braided stainless-steel wire	Silicone
Spectra (discontinued, replaced by Tactra)	Boston Scientific	Mechanical : articulating high-molecular-weight polyethylene and titanium segments	—

- All three current rods have trimmable proximal tips or segments (Tactra and Genesis with end caps) and come in **fixed girths**.
- The examinable distinction is the core metal: nitinol (Tactra), silver (Genesis), stainless steel (Rigi 10).

TABLE 75.1 SIZES AND COMPONENTS OF PENILE IMPLANTS

Vendor / device	Expansion	Lengths (cm)	Diameter or maximum expansion (mm)	Construction
Boston Scientific AMS 700 LGX	Girth + length	12, 15, 18, 21	18	Trilaminate: silicone outer, polyester/spandex middle, silicone inner
AMS 700 CX	Girth	12–24 (steps of 3)	18	As LGX
AMS 700 CXR	Girth, narrow	10–18 (even)	14	As LGX
Tactra	Malleable	14–23, 16–25, 18–27	9.5, 11, 13	Silicone over nitinol
Coloplast Titan / Titan Touch	Girth	14–28 (even)	19.2–22.1, rising with length	Polyurethane (Bioflex)
Titan Narrow	Girth, narrow	11, 14, 16, 18	15.2–17.9, rising with length	Polyurethane (Bioflex)
Genesis	Malleable	14–23, 16–25, 18–27	9.5, 11, 13	Silicone over braided silver
Rigicon Infla 10 AX	Girth + length	12, 15, 18, 20, 22, 24	18–23, rising with length	Silicone / spandex-polyester / silicone; silicone spray at input tube
Infla 10 X	Girth	12, 15, 18, 20, 22, 24	18–24.5, rising with length	As AX
Infla 10 NB	Girth, narrow	10, 12, 14, 16	14.5	As AX
Rigi 10	Malleable	14–23 (9, 10 mm); 16.5–25 (11, 12, 13 mm)	9, 10, 11, 12, 13	Silicone over braided stainless steel

- **What the table is for:** matching device to corporal size and to the problem — **only two cylinders expand in length (LGX, Infla 10 AX); each vendor has one narrow cylinder (CXR, Titan Narrow, Infla 10 NB) for small or scarred corpora.**
- **Pattern:** Coloplast alone uses **polyurethane (Bioflex)**; Boston and Rigicon use a **silicone–fabric–silicone trilaminate** — this underlies differences in **cautery tolerance** and in **how cylinders fail** (fabric shredding versus polyurethane deterioration).
- **Trap:** in Coloplast and Rigicon inflatables **girth expansion increases with cylinder length**; Boston cylinders have one fixed maximum per model.

Concept — Every device choice trades **hydraulic rigidity-with-flaccidity** against the **simplicity and near-zero mechanical failure of a rod**; within inflatables, the choice is **girth-only versus length-expanding and standard versus narrow**.

INFORMED CONSENT

- A thorough explanation of procedure and expected result is paramount before implantation.
- The greatest postoperative disappointment is an erection shorter than the patient remembers — caused by **pseudocapsule scar forming around every part of the device and limiting cylinder expansion**.
- **Apparent size reduction is greater in Peyronie disease**, where tunical scar already limits expansion. **Stress this potential.**
- **Devices are mechanically reliable but not indestructible** — future repair may be needed.

- **Tissue problems — pump migration, erosion and extrusion of parts, infection** — may all require reoperation. **Sensation and climax can be affected.**
- **Alternative ED treatments must be explained and offered**, although usually already tried.
- **Consent tends to be more extensive with more experienced implanters.**
- Implant candidates are often **highly focused on their genitalia; disappointed patients can be litigious.**
- **Unrealistic expectations are common:** confirm that the planned outcome is **a firm penis usable for penetrative intercourse, not necessarily with the feelings, desires and abilities known previously.**

KEY POINTS: INFORMED CONSENT FOR PENILE IMPLANT PLACEMENT

- **The erect penis may be shorter and thinner** than preoperatively.
- **The goal is a firm penis suitable for intercourse.**
- **Penile sensation and ejaculation may both be diminished.**
- **Varying degrees of postoperative pain.**
- **Reoperation for mechanical problems** may be necessary.
- **Tissue problems such as infection** may necessitate further surgery.
- **Other ED treatments exist:** oral medications, penile inserts and injections, VEDs.
- **A variety of implants exist**, including three-piece inflatable and malleable semirigid rods.

Concept — Consent is **expectation management**: the device restores rigidity, not the former erection, and length loss is the complaint to pre-empt.

PREOPERATIVE PREPARATION

- **Examine genital and pelvic skin: phimosis, sebaceous lesions, fungal infection** must be treated and eradicated first.
- **Phimosis: correct first, implant about 6 weeks later** once healed.
- **Redundant foreskin with a wish for circumcision: perform simultaneously — implant first, close all incisions, then circumcise.**
- **Strong-soap washing at home** is reinforced.
- **Shave in the preoperative area or theatre**, so skin nicks cannot become colonised before surgical preparation.
- **Diabetes mellitus:** glycaemic targets in the table below.
- **Urine must be sterile. Stop anticoagulants, including low-dose aspirin.**
- **Ascertain allergies, especially to antibiotics** — used systemically, in irrigating or dipping solutions, and as implant coatings.
- **Obtain insurer approval** (costly; some policies exclude sexual dysfunction); **discounted cash arrangements** may be available.

GLYCAEMIC CONTROL AND IMPLANT INFECTION

Acute (day of surgery) — decisive	Chronic control
Blood glucose under 200 mg/dL at the time of surgery; if raised that morning → intravenous line and hourly 10-unit insulin boluses	Long-term control (glycated haemoglobin, HbA1c) is less important in avoiding implant infection

Concept — Preparation is **infection prevention by source control**: skin, urine and acute hyperglycaemia are each corrected before the foreign body goes in.

INCISIONS

SURGICAL APPROACHES COMPARED

Approach	Access and technique	Advantages	Cautions / status
Infrapubic (original Scott)	Vertical from penile base toward umbilicus (transverse variant); dorsal corporal surfaces for cylinders; linea alba to space of Retzius for reservoir	Direct reservoir placement under vision	Dorsal neurovascular bundle at risk — stay sutures lateral to it at 10 and 2 o'clock ; bundle nerves are sensory to glans and distal shaft ; fundiform ligament (from mid-symphysis) marks the midline
Penoscrotal	Originally on the scrotal median raphe ; transverse upper scrotal variant gives better exposure of proximal crura and external inguinal ring; Foley catheter placed first to locate corpus spongiosum	Secure pump placement in a subdartos pouch ; more cosmetic incision	Used by the majority today for three-piece devices
Subcoronal degloving for three-piece	Cylinders via penoscrotal or infrapubic corporotomies	Shaft reconstruction for straightening easily performed	Recent report
Subcoronal for semirigid rods	Lateral corporotomies	Usual rod approach	Long proximal dilation (difficult with scar); little tissue for layered closure ; circumcise uncircumcised patients simultaneously — else maceration of suture line or lymphoedema of remnant foreskin
Ventral penile	Skin incision on proximal shaft over the spongiosum ; vein retractor draws foreskin distally to reach corporotomy sites as in subcoronal; corporotomy end lifted over rod tip	No need to bend the rod ; skin and corporotomy incisions do not overlap ; foreskin retained	Alternative to subcoronal for rods
Perineal (Small-Carrion)	—	—	No longer used

- **What the table tests:** which approach suits which device — **penoscrotal for three-piece** (pump security), **subcoronal or ventral for rods**.
- **Trap:** the infrapubic approach is the one that endangers the **dorsal nerve**; the penoscrotal approach endangers the **urethra**, hence the Foley catheter.

Concept — *Incision choice trades **reservoir exposure** against **structure at risk**: infrapubic threatens the dorsal nerve, penoscrotal the urethra — and the penoscrotal route wins because it anchors the pump.*

TECHNIQUE OF DEVICE PLACEMENT

Corporal Exposure, Dilation and Sizing

Supine → **catheter drains bladder** (safe reservoir placement, avoids urethral injury) → **expose tunica**; skeletonise corpora **lateral to urethra** (penoscrotal) or **lateral to neurovascular bundles** (infrapubic) → **two 2-0 polydioxanone (PDS) horizontal mattress stay sutures about 1 cm apart** on each corpus (retraction during dilation; later closure) → **longitudinal corporotomy** between stays by cautery or scalpel, parallel to shaft, extended as needed → **dilate proximal then distal**: Metzenbaum scissors, then **Hegar or**

Brooks dilators 7 → 13 mm → goalpost test → measure → size and place cylinders → test inflation 1 → close corporotomies → test inflation 2 → pump → tubing connection → test inflation 3 → irrigate and close

- **Initial corporal tunnelling is paramount** to avoid crossover, crural perforation and urethral injury.
- **Proximally:** pass dilators **gently to the ischial tuberosity** — perforation risk at the crural insertion.
- **Distally:** instrument points directed laterally, especially near the glans → avoid **urethra** and the **thin midline septum**.
- **Goalpost test:** dilators simultaneously in both corpora, **proximally then distally**; check **symmetry**.
- **Fibrosis** (notably **post-priapism ED**) → **cavernotomes** to open or remove scar.
- **Suspected distal perforation** → **irrigate into the corporotomy** and look for **leak near the urethral meatus**.
- **Measurement:** **Furlow device, stay suture as reference; proximal + distal = total corporal length** (illustrated with an 11-cm proximal reading).

CYLINDER LENGTH SELECTION

Inflatable cylinders	Semirigid rods
Equal to, or 1 cm shorter than, measured length — avoids intracorporal buckling and S-shaped deformity ; some prefer slightly longer to stretch the tunica over time	Downsize 0.5–1 cm from total corporal length
Length adjusted with rear tip extenders	Longer corporotomy if midcorporal; proximal crural end placed first
Placed with Furlow introducer ; Keith needle delivers traction suture through the glans	Vein retractor elevates distal corporotomy for subcoronal placement

- **Test inflation 1 (before closure):** seating, crossover, curvature, inadvertent perforation.
- **Minimal-dilation technique:** in a **primary implant with little or no corporal scar**, pass only the measuring tool, then place cylinders without further dilation → **greater penile length and less pain** than standard dilation.

Closure, Pump and Connection

- **Close corporotomies over the cylinders with the stay sutures**, or with interrupted/running sutures, taking **extreme care not to puncture the device**.
- **Protect the cylinder** with a small malleable or proximal pushing instrument; **longitudinal traction via the glans suture** minimises protrusion and puncture.
- **Keep input tubing at the most proximal point of the corporotomy**.
- **Test inflation 2 after closure**.
- **Pump:** subdartos pouch in the **most dependent scrotum**, right or left (most patients manage either), **anterior and lateral to the testis**, by finger, ring clamp or nasal speculum dissection.
- **Shodded occlusive clamps; trim all redundant tubing; make a straight pump-to-reservoir connection**.
- **Test inflation 3:** malfunction, displacement, fluid leak, adequate sizing.
- **Irrigate thoroughly; close with two subcutaneous layers of 3-0 Vicryl and a 4-0 Monocryl skin layer**.

Girth Selection for Malleable Rods

- **Rods have fixed girths** — choose width **before opening packages**: place **two Hegar dilators parallel in both corpora**, then try to **touch thumb to index finger between them** (technique applicable to Ambicor, Tactra and Genesis).

TWO-DILATOR GIRTH TEST

Finding	Interpretation for same-girth cylinders
No separation of dilators	Tight fit
Slight separation	Ideal fit
Thumb touches index finger	Loose fit

Concept — Placement is governed by **three test inflations and lateral instrument direction**: every step is designed to detect crossover, perforation or malposition before the wound closes.

Reservoir Placement**RESERVOIRS BY VENDOR**

Vendor	Shape	Volumes	Lockout valve
Boston Scientific	Spherical; flat disc 'Conceal'	Spherical 65 and 100 mL; Conceal 100 mL	On the pump
Coloplast	Cloverleaf (flat)	75 and 125 mL	At tubing–reservoir junction
Rigicon Infla 10	Oblong spherical	65, 75 and 100 mL	At tubing–reservoir junction

- **Lockout valves prevent spontaneous transfer** of fluid from a pressurised reservoir to the cylinders (autoinflation) — what makes ectopic submuscular placement viable.
- **Flat reservoirs (Conceal, cloverleaf)** rest easily in the submuscular space.

Space of Retzius

- **Original 1970s route**: inferior linea alba, between the recti, into the prevesical space of Retzius — still simple and rapid.
- **Via penoscrotal incision**, through the inguinal canal to the side of the bladder:

Drain bladder → Trendelenburg (bowel off pelvis) → sharp instrument pointing medially pierces transversalis fascia just above pubic tubercle → index finger confirms space of Retzius → long nasal speculum beside finger → finger withdrawn; reservoir inserted through speculum → fill with isotonic saline — expansion enlarges the cavity

- Iliac vessels lie about 1 inch lateral to the reservoir; if resistance is met, choose another location.
- Prior radiation, previous perivesical urinoma, previous surgery → adhesions; disrupting them may injure iliac vessels.
- Large reservoirs can compress the iliac vein → leg oedema.
- After robotic radical prostatectomy the peritoneal reflection is not replaced → bowel migrates to the internal ring and becomes fixed by adhesions; inguinal Retzius placement has then compressed bowel on inflation, requiring surgical intervention.

Ectopic Locations

- **Submuscular suprafascial (commonest ectopic site)**: reservoir between rectus muscle above and transversalis fascia below.

Finger through inguinal canal creates space above the fascia toward the umbilicus → long nasal speculum → long (double-hinged) clamp develops cavity 3–4 inches inside internal ring → reservoir nipple grasped, driven in, held while filled with saline

- **Visible in thin patients** — discuss preoperatively; some report initial discomfort.
- **Confirm location after placement** — reservoirs have been found intraperitoneally.

- **Fill almost to capacity:** the tendency to fill only slightly beyond cylinder volume (more comfortable, less visible) lets the reservoir **migrate out of the inguinal canal during inflation**.
- **Revising a submuscular reservoir:** inflate it, cut directly over it, remove and replace through that incision — **not** by incising scar along the tubing as for Retzius reservoirs (muscle destruction, significant bleeding).
- **Decommissioning:** drain and leave the old reservoir, place a new one elsewhere.

RESERVOIR LOCATIONS COMPARED

Site	Access	Indication	Key caution
Space of Retzius	Linea alba (infrapubic) or through inguinal canal (penoscrotal)	Standard, unscarred pelvis	Iliac vessels 1 inch lateral; adhesions; bowel after robotic prostatectomy
Submuscular suprafascial	Through inguinal canal, above transversalis, below rectus	Commonest ectopic site	Visibility; intraperitoneal misplacement; underfilling → migration
Lateral retroperitoneal	Separate incision 1 inch medial to anterior superior iliac spine	After cystectomy and urinary diversion	Tubing tunnelled to pump site
Epigastric	Left subcostal; external oblique aponeurosis incised, internal oblique and transversus split; above omentum	Reservoir completely out of the pelvis	Seldom used; tubing tunnelled subcutaneously to scrotum
Fat above rectus fascia	Subcutaneous	Very obese patient	—

- Ectopic sites are especially suited to significant intra-abdominal scarring — after cystectomy with urinary diversion or with a renal transplant.
- **Lateral retroperitoneal sequence:** incise external oblique aponeurosis parallel to fibres → develop fatty extraperitoneal space bluntly → place and fill → close aponeurosis with tubing exiting inferiorly → tonsil clamp from infrapubic or penoscrotal incision retrieves tubing to the connection site.

KEY POINTS: TECHNIQUE OF DEVICE PLACEMENT

- Three-piece inflatables give the best rigidity and flaccidity and are by far the most popular with patients and urologists.
- A shorter erection is the major reason for disappointment — stress and document it at consent.
- Limited dexterity → encourage a semirigid rod, avoiding frustration with a hydraulic pump.

Concept — Reservoir safety rests on one rule — **a capacious, unscarred space**; the lockout valve is what allows the reservoir to leave the pelvis for such a space when Retzius is hostile.

INTRAOPERATIVE TROUBLESHOOTING

Fibrotic Corpora

- Dilation is usually straightforward.
- Pelvic radiation (common in prostate cancer, frequent implant recipients) does not increase placement difficulty or postoperative complications.
- True scar: after acute ischaemic priapism or after explantation for infection or erosion — creating cavities is difficult.
- Midshaft corporotomy lets dilators travel an equal distance each way.

- **Use the opposite incision to the original** (penoscrotal ↔ infrapubic) → less scar encountered.
- **Instruments always point dorsolaterally away from the spongiosum, parallel to the shaft, and are seen moving against the lateral shaft wall.**
- **Scissors: spreading, not cutting.**
- Once a channel reaches the **distal subglandular area** and the **proximal ischial tuberosity**, broaden it with cavernotomes or the Otis urethrotome.

INSTRUMENTS FOR SCARRED CORPORA

Instrument	Action
Rossello cavernotome (Coloplast)	Raised metal cups shave tissue like a wood rasp; rod pulled in and out
Uramix cavernotome (Lansdowne, PA)	Raised metal blade shaves as the instrument is oscillated side to side
Otis urethrotome	Cuts in one plane; each cut allows the next dilator size; blade pointed away from the urethra and rotated slightly between cuts to avoid cutting through the tunica
Small sharp knife blade	Excises blocks of scar; especially useful for scar exposed at the corporotomy

- **Narrow inflatable cylinders (all three vendors)** fit a scar-restricted cavity and give **rigidity comparable to standard cylinders**, because they **completely fill the chamber**.
- **A single cylinder gives suboptimal results — lopsided erection, compromised rigidity.**

Cavernosal Reconstruction

Incise scarred chamber deeply along its full length → lay cylinder in the cavity → cover with a synthetic or natural tunical substitute

TUNICAL SUBSTITUTES FOR CAVERNOSAL RECONSTRUCTION

Class	Materials	Fixation	Notes
Biologic (preferred)	Cadaveric pericardium (Tutoplast); porcine small intestinal submucosa (Surgisis)	Absorbable suture	Shrink with time — oversize accordingly; form a lattice on which native scar forms over 3 months; lower infection rate
Synthetic	Polyethylene terephthalate (Dacron); polytetrafluoroethylene (Gore-Tex)	Soft nonabsorbable sutures	Graft same size as defect
Collagen fibrin patch	TachoSil — equine collagen matrix with fibrin sealant, marketed for vascular surgery	No sutures; held in place briefly	Overlap tunical edges by about 0.5 cm

- **Pair to remember: biologic grafts shrink (oversize them, absorbable suture); synthetic grafts do not (size to defect, nonabsorbable suture).**
- **Cavernosal reconstruction is now infrequent** — simpler cavity-creating techniques have improved.

Concept — In scarred corpora the escalation runs **direction → enlargement → narrow cylinder → reconstruction**; every step preserves two cylinders, because a single cylinder fails.

Crural Crossover

- **The septum is thin and easily breached.** If both cylinders lie on one side: **lopsided erection, second cylinder difficult to place, corporal measurements markedly different.**
- **Recognise and correct intraoperatively**, not when the patient reports an odd-looking erection.

Remove both cylinders → large dilator in the common cavity → redilate the opposite side with instruments pointing laterally; place a cylinder → remove the dilator → second cylinder into the original common cavity

- **Confirm:** inflate — **Foley catheter midline with cylinders symmetrical either side.**
- **Goalpost test** for proximal or distal crossover: **a dilator in each crus, parallel and of equal height.**

Corporal Perforation

- **Unusual; more common at secondary insertion**, where scar forces aggressive dilation.
- **Proximal (crural) perforation suspected** when **one dilator passes further inferiorly than its mate** and/or **corporal measurements differ by more than 1 cm**; there is **no support**, and at the extreme the **rod is palpable 2 inches lateral to the anus.**
- **Former repair — windsock:** vascular graft cup as a rear tip extender sewn to the tunica; **abandoned** — infection source and difficult removal because of tissue ingrowth.
- **Current repair — suture sling:**

Double-armed 2-0 Prolene (nonabsorbable polypropylene) through **proximal cylinder end or rear tip extender** → cylinder placed; **distal insertion string pulled to level with its mate** → **partial inflation** → **suture ends brought out each side of the tunica just distal to the input tube exit** → corporotomy closed → **sling tied firmly over the closure; knot cut long** for identification at future replacement

PERFORATION SITES AND MANAGEMENT

Site	Recognition	Management
Proximal crural	Dilator deeper than mate; lengths differ by more than 1 cm ; no support	Suture sling
Urethra at the corporotomy	Direct view	Accessible → close securely, irrigate copiously, place cylinder; any doubt about closure → abort that side to heal
Distal — usually fossa navicularis	Blood at meatus, dilator visible at meatus, irrigant exiting urethra	Do not place the ipsilateral cylinder; no catheter needed; place contralateral cylinder to maintain length; replace ipsilateral at 5–6 months

- **Distal defects are not closed** — poor exposure.
- **No urethral catheter:** the urinary stream **draws tissue fluid into it (Bernoulli principle)**, so urine does not enter the defect.
- **Delay of 5–6 months** allows solid scar to form, so the dilator does not re-enter the old tract.
- **Alternative:** create a **hypospadias**, repair the perforation securely, place **both cylinders** — successful in their small series, but **half later requested hypospadias repair for spraying.**
- **Never assume a urethral injury will heal without closure; a cylinder left beside an unclosed injury will likely become infected.**

Hypermobile Glans Penis

- **Glans floppy despite correctly placed cylinder tips** → penetration difficult; **dorsal, ventral or both;** commoner in uncircumcised men.
- **Ventral hypermobility = SST deformity** — resembles the **drooping nose of the supersonic transport aircraft.**
- **Mild:** tense the foreskin toward the **scrotum during penetration**, seating the glans on the cylinders.
- **Otherwise:** glans fixation to the distal tunica.

Unidirectional: hemicircumcising incision → **partially dissect glans off distal tunica** either side of the **spongiosum** (ventral) or **dorsal neurovascular bundle** (dorsal) → **nonabsorbable 3-0 soft suture on an SH needle** through glans substance, then tunica **at the level of the inflated cylinder tip** → **tunica lifted in Adson forceps teeth** to avoid impaling the cylinder → **place both sutures before tying** (exposure) → **two-layer closure**

- **Excessive mobility in both directions** → **four-quadrant fixation through a complete circumcising incision.**
- **Glans hypermobility and short cylinders are distinct problems:** short cylinders → longer cylinders or rear tip extenders, not glans fixation.

Curved Erection After Cylinder Placement

- **Usually straight** — the tunica expands uniformly.
- **Tunical scar from Peyronie disease, a prolonged pharmacological injection programme, or pelvic trauma** → **curvature toward the scar.**
- **Semirigid rods can usually be bent** to overcome it; **extensive thick scar** needs tunical manoeuvres.
- **Cylinder choice (table below):** one of the three high-rigidity cylinders alone straightens most cases.
- **Residual curvature under 20 degrees is acceptable** for penetration.

Modelling

Shod-clamp pump tubing (protects pump from back pressure) → **bend inflated cylinders forcefully to a right angle away from the curve** → **hold 90 seconds** → **repeat if needed**

- **Check for corporotomy disruption and urethral injury** after modelling.
- **Works best for untreated Peyronie plaques; less successful for scarring after explantation.**
- **Keep the penis partially erect for 3 months** so the pseudocapsule forms straight; **deflated cylinders allow the curve to return.**

Recalcitrant Curvature

- **Nesbit procedure, tunical plication, or plaque incision and grafting.**
- **Nesbit:** excise small tunical ellipses from the **convex side** and close with absorbable suture; or **plication stitches** limiting convex expansion.
- **Incision through scar on the concave side** lets the cylinder force straighten the erection; **graft the defect if necessary** — **defects under 1 cm need no graft.**
- **Cautery over Titan (Bioflex) cylinders:** below 35 watts; **Boston Scientific silicone is unaffected by cautery.**

CYLINDER CHOICE FOR CURVATURE

Better correctors (intrinsic rigidity)	Poorer correctors (distally expanding)
AMS 700 CX, Titan, Infla 10 X	AMS 700 LGX, Infla 10 AX

KEY POINTS: INTRAOPERATIVE TROUBLESHOOTING

- **Injuries between the corpus cavernosum and the exterior must be closed securely** before a cylinder is left in place.
- **AMS 700 CX, Titan and Infla 10 X** have the best intrinsic rigidity and are better suited to correcting curvature from tunical scarring.

- **During dilation, scissors and dilators point dorsolaterally and parallel to the shaft** to avoid crural crossover and urethral injury.

Concept — Almost every intraoperative complication — crossover, perforation, urethral injury — is prevented by instrument direction and detected by symmetry; the goalpost test and the midline Foley catheter are the checks.

POSTOPERATIVE COMPLICATIONS

Implant Infection

- **Incidence low (1–3%)**, but infection still occurs; precautions span **before, during and after** surgery.

Antibiotic Prophylaxis

- **Systemic prophylaxis recommended by American Urological Association (AUA) guidelines.**
- **Vancomycin or a third-generation cephalosporin, plus gentamicin, intravenously 1 hour before incision, continued up to 24 hours or until the wound is sealed.**
- **No antibiotic is needed beyond that, yet most urologists give 5–14 days of oral antibacterial; explantation for infection is the same with or without postoperative antibiotics, which may even be harmful.**
- **Organisms from infected implants (multi-institutional series): most, not all, sensitive to vancomycin–gentamicin; vancomycin + an antifungal (amphotericin or fluconazole) + piperacillin–tazobactam would cover all → prudent prophylactically in high-risk patients and in documented infection before cultures return.**
- **Caution: nephrotoxicity with combined vancomycin–piperacillin.**
- **Vancomycin–gentamicin alone is associated with higher infection risk than non-standard regimens; future guidelines will clarify.**

Irrigation, Coatings and Technique

- **Copious irrigation throughout with anti-infective solution or saline; no particular antiseptic is currently recommended; chlorhexidine gluconate (Irrisept) is gaining popularity.**

ANTIBIOTIC COATINGS OF CURRENT IMPLANTS

Coating	Devices	Agent
Inhibizone (impregnated)	AMS 700 series	Rifampin–minocycline — chosen for low allergic potential and activity against implant organisms
Hydrophilic (adsorbs dipping solution)	Titan, Genesis, Infla 10, Rigi 10	Surgeon's dip: rifampin–gentamicin or sulfamethoxazole–trimethoprim

- **All three vendors' three-piece devices are coated; the drug recedes into tissues over several days.**
- **These modifications have halved infection rates.**
- **Eid's no-touch technique:** skin completely draped off; any glove or instrument touching skin is discarded → infection rate under 0.5% in more than 1500 cases. Implication: organisms enter through the open wound, not the bloodstream.
- **Chlorhexidine–alcohol is the best skin preparation** — about 40% lower surgical site infection than povidone-iodine (randomised trial).
- ***Staphylococcus aureus* is carried by about a quarter of the population, most densely in the nasal cavity; methicillin-resistant *S. aureus* (MRSA) carriage is a small fraction of this.**

- **MRSA screening:** most prosthetic urologists do not screen; **American College of Surgeons** guidelines recommend it for **cardiac and orthopaedic joint surgery**; prosthetic urology will likely be added.
- **Positive nasal MRSA → decolonisation:** **chlorhexidine bathing on days 1, 3 and 5 preoperatively** plus **2% nasal mupirocin (Bactroban) twice daily for 5 days**.

Diagnosis and Management

- **Suggestive:** fever, chills, erythema at the incision.
- **Definitive:** **purulent drainage (especially on squeezing cylinders or pump)**, **visible prosthetic material**, **progressive pain**, **pump fixed to the scrotal wall**.
- **Infection present → open the incision and remove ALL foreign material;** a retained component risks abscess around it.
- **Discuss both options with the patient before operating.**

OPTIONS AFTER REMOVING AN INFECTED IMPLANT

Option	Technique	Outcome / caveats
Explant with delayed replacement	Irrigation drain in each corpus, wound closed; organism-specific antibiotic solution every 8 hours for 3 days (clears microbes, prevents corporal abscess); return at 2–6 months	Difficult reinsertion from corporal scarring; noticeably shorter erection — delayed reimplantation loses several centimetres of corporal length ; VED 10 minutes daily while waiting limits shortening and keeps the erection straight
Mulcahy salvage (rescue) — original protocol	Remove all foreign material → wash with antiseptic series → change drapes → immediate new implant ; originally pressure washing with antibiotics, povidone-iodine and hydrogen peroxide	Success 84%; failures: onset within 2 months of implantation, aggressive organisms, purulence with cellulitis
Delayed salvage for aggressive infection	Systemic vancomycin + gentamicin , watch cellulitis recede, salvage at 3 days	Cleared all such infections in a small series
Salvage with semirigid rods	Rods rather than the original device type	High success ; avoids parts in a macerated scrotum or remote abdomen; quicker; about one-third later converted to three-piece

- **What the table is for:** salvage techniques exist to **preserve corporal length**, which delayed replacement sacrifices.
- **Trap:** salvage failure predictors — **early onset, aggressive organism, purulence with cellulitis**.

BOX 75.1 ANTISEPTIC WASHES USED DURING A SALVAGE PROCEDURE

Step	Current recommendation
1	Dilute povidone-iodine (0.35–3.5%) in implant cavities, left 3 minutes
2	Wash out with saline or antibiotic solution
3	Eliminate hydrogen peroxide — no antiseptic advantage over povidone-iodine and cytotoxic
4	Chlorhexidine may be more effective than povidone-iodine, but data insufficient
5	Pressure irrigation is optional

- **Basis:** the original solutions were empirical; **orthopaedic joint-infection literature** reshaped them — **dilute povidone-iodine beats full or half strength**; **chlorhexidine gluconate needs more study**.

Biofilm and the Revision Washout

- **Repair of a clinically uninfected malfunctioning implant carries about 10% infection.**

- **Mechanism:** organisms introduced at the original operation form a **mucopolysaccharide biofilm** shielding them from host defence; **repair disturbs it → bacteria activate → symptomatic infection.**
- **Revision washout:** antiseptic irrigation series in each cavity after old parts are removed and before new parts are placed → infection reduced to about 3%. Recommended for all secondary procedures in which old parts are removed.

Concept — Infection is a **biofilm on a foreign body introduced through the wound**: prevention targets the operative field, and cure demands removal of every component — salvage merely shortens the interval before a new device.

Cylinder Aneurysm

BULGE AT THE BASE OF THE PENIS: EARLY VERSUS LATE

Early postoperative	Years after implantation
Disrupted corporotomy with cylinder herniation	Weakened cylinder wall from frequent maximal inflation and aggressive use
Explore and repair the corporotomy; graft usually unnecessary	Replace the cylinders; tunica may need tapering; graft usually unnecessary

- **Diagnosis:** magnetic resonance imaging (MRI) identifies it, but experienced examining fingers usually suffice.
- **Failure mode by device:** AMS 700 middle fabric layer shreds; Titan polyurethane deteriorates with excessive wear.
- **Disrupted AMS 700 outer silicone** → tissue ingrowth into the spandex middle layer → difficult removal needing sharp and blunt dissection and aggressive tearing; small retained fabric fragments are harmless.

Cylinder Extrusion

- **Cylinder tip extrudes through the tunica at the end of the chamber** — palpable under the foreskin at the side of the penis, or **protruding into the glans**, sometimes as **impending erosion into the urethra at the meatus.**
- **Causes of tunical wear:** forceful dilation, oversized cylinders, aggressive use.
- **Not exposed and not infected** → **distal corporoplasty:** reseat the tip in a new plane where distal tunica is intact.

Hemicircumcising incision → longitudinal corporotomy over the extruded tip → tip delivered; traction suture through its end retracts cylinder inferiorly → incise the back wall of the pseudocapsule → develop a dorsomedial plane in spongy erectile tissue with scissors and sequential dilators → traction suture guides cylinder into the new cavity → close corporotomy

- **Oversized cylinder** → replace with an appropriately downsized device.

Cylinder Erosion

- **Chronic tip wear** → extrusion beyond the tunica through skin or into the urethra.
- **Associations:** underlying device infection; excessive intracorporal pressure (e.g., crural crossover); external friction from repeated catheterisation or instrumentation.
- **Chronic catheterisation, indwelling or intermittent,** rubs the tip in the fossa navicularis → **high incidence of urethral erosion.**
- **Semirigid rods in insensate patients (e.g., spinal cord injury)** erode more: rod tips are hard and cannot be recessed, and the firm penis rubs on undergarments → **use hydraulic implants in these patients.**

- **Urethral erosion sites cannot be repaired** → remove cylinders and all other parts, antiseptic washout (organisms travel along tubing to other components).
- **Spacer cylinder in the opposite corpus** maintains patency and length; **replace the eroded side at 5–6 months** once solid scar obliterates the tract.
- **Lateral erosion is treated exactly as urethral erosion.**
- **Immediate new cylinder with antiseptic irrigation and distal corporoplasty after erosion usually fails.**

EXTRUSION VERSUS EROSION

Extrusion	Erosion
Tip through the tunica , still covered by skin/glans	Tip through skin or into the urethra — exposed, contaminated
Uninfected	Contaminated ; may reflect device infection
Distal corporoplasty — device retained, tip resealed	Remove entire device , washout, contralateral spacer, replace at 5–6 months

Visceral Erosion of the Reservoir

BLADDER VERSUS BOWEL EROSION

Bladder	Bowel
Very rare ; commoner of the two	Rarer than bladder
Restricted perivesical location : constant balloon pressure breaks down the wall, or wall torn during expansion in a scarred space	Reservoir adjacent to an immobile bowel segment fixed by adhesions ; expansion tears the wall
Scar forms quickly; at cystotomy tubing exits a small hole in bladder-wall scar	Quickly evident — succus entericus invades tissues, marked inflammation
Wound and urine usually sterile → remove reservoir, stitch the exit site, close bladder, new reservoir in a capacious site ; antiseptic irrigation	Remove the entire implant ; general surgical consultation for bowel

- **The contrast is sterility**: bladder erosion → **reservoir exchange only**; bowel erosion → **whole device out**.

Penile Necrosis After Implant Placement

- **The penis is a digit**: nutrient flow runs **proximal to distal**; ischaemia → **dry gangrene**; the **glans, most distal**, is most susceptible.
- **Longitudinal incisions (foreskin and tunica, parallel to shaft)** interrupt distal supply less than transverse incisions.
- **At-risk patients**: severe cardiovascular disease, diabetes mellitus, chronic smoking, previous prosthetic implants, prior pelvic radiation — especially two or more.
- **Distal foreskin nourishes the glans**; circumcising incisions may contribute to glans necrosis.
- **Tunical incisions to straighten or lengthen** → caution in high-risk patients.
- **Mummy wrap without excessive pressure**; always leave the **glans exposed** to monitor perfusion.
- **Necrosis detected** → **remove cylinders promptly** to relieve pressure; **conservative debridement** aids later reconstruction.

KEY POINTS: POSTOPERATIVE COMPLICATIONS

- **In implant infection, remove all foreign material** — remnant parts harbour microbes that perpetuate infection.

- **Patients at risk of reduced penile perfusion** should be counselled about distal necrosis; **limit or stage aggressive reconstruction**.
- **Place the reservoir in a capacious cavity** to avoid injuring bowel or bladder during filling and expansion.

Concept — *Late complications share one mechanism — **sustained pressure or friction on tissue that cannot yield** (tunica, urethra, bladder, glans perfusion); management depends on whether the device has become **contaminated**.*

SPECIAL SITUATIONS

Immunosuppressed Patients

- **Successful implantation after renal, cardiac, hepatic and renal-pancreas transplantation** — **no increase in complications, including infection**.
- **Transplant specialist manages immunosuppression**; prophylactic antibiotics coordinated with the urologist.
- **Human immunodeficiency virus (HIV)-positive men on antiviral therapy with appropriate cluster of differentiation 4 (CD4) counts: no increased infection**.

Penile Implants in Ischemic Priapism

- **Acute ischaemic priapism lasting more than 36 hours** → **completely scarred corpora**; months later an implant is the **only option** — cavities difficult, **erection significantly shorter**.

TIMING OF IMPLANTATION AFTER ISCHAEMIC PRIAPISM

Immediate (during the acute episode)	Delayed (months later)
Recent trend	Traditional
Corporotomy relieves compartment-syndrome pain	No symptomatic benefit
Oedematous cavities, no solid scar → most erectile length preserved	Dense fibrosis → difficult dilation, significant shortening

- **Hydraulic or semirigid cylinders** may be used.
- **Document spongy tissue necrosis (e.g., biopsy)** — medicolegal protection.
- **Distal corporal shunts are likely** → **close the defects and avoid distally expanding cylinders (AMS 700 LGX, Infla 10 AX)**.
- **Consent is paramount**: usually **young men without prior ED** learning that natural erection is lost; **counsel increased infection and size-loss risk**.

Lax Suspensory Ligament

- **Suspensory ligament attaches the crura to the symphysis pubis and ischiopubic rami**; supports the erection and **prevents proximal migration during thrusting**.
- **Infrequent laxity**: cause unclear — **congenital or sexual trauma**.
- **Well-sized, well-filled cylinders still buckle** (poor axial rigidity); examination shows a **gap or cleft between crura and bone** and a penis that **moves easily side to side**.
- **Repair**: two or three nonabsorbable interrupted sutures fixing the tunica to the periosteum of the symphysis and ischiopubic ramus bilaterally.

Old Biofilm in Long-Indwelling Implants

- **Very rarely, calcified debris** is found when repairing implants **in place for over 10 years** (illustrated around a 15-year-old Titan).
- **Cultures negative**; microscopy: **amorphous material with calcification** — likely **remnant calcified biofilm**.
- **Wash out as completely as possible and replace parts** — nothing more needed.

Penile Implants After Phalloplasty

- **Phalloplasty in transgender men (female-to-male)** is increasingly common.
- **Commonest neophallus: radial forearm flap tubularised into a neourethra** with surrounding skin and fat.
- **Implant placed after the graft has matured; single cylinder, usually Titan or AMS 700 CX, proximal end anchored to the symphysis pubis or ischiopubic ramus.**
- **High complication rate, especially erosion** — **no native tunica albuginea**, the **second toughest membrane in the body after the dura mater**, to contain the cylinder.

Long-Distance Bicycle Riding

- **Crura containing proximal cylinder ends or rear tip extenders abut the anterior horn of a standard saddle.**
- **Short recreational riding: no problem. Frequent long-distance riding → pain and/or numbness; also associated with urethral stricture.**
- **Solution: a saddle with buttock supports only (no nose)** — also used by riders with an **artificial urinary sphincter**.

Concept — Each special situation is about the **cylinder's container** — *scarred corpora, an absent tunica, a lax ligament, a compressed crus* — and management restores or substitutes that support.

POSTOPERATIVE CARE

- **Most patients go home the same day.**
- **Leave inflatable cylinders partially erect** → dilated corpora heal with scar around inflated cylinders, preventing a contracted **pseudocapsule** that would limit expansion.
- **Compressive mummy wrap** reduces bruising and oedema.
- **Short-term scrotal drains:** prevent haematoma, **not proven to raise infection risk; remove within 24 hours or when bleeding stops.**
- **Compressive underwear for 4–6 weeks, penis pointed up toward the umbilicus** → avoids permanent downward angulation.
- **Indirect ice to the scrotum** as needed.
- **Oral opioid analgesia for about a week, then nonsteroidal anti-inflammatory drugs.**
- **Dorsal neurovascular bundle infiltration with lidocaine and bupivacaine** reduces pain and analgesic need; **long-acting liposomal bupivacaine (Exparel)** cut analgesic use **threefold** versus standard bundle block.
- **Infrapubic approach: patient pulls the pump into a dependent scrotal position daily** before activation; **penoscrotal approach: pump secured in the dartos pouch** — unnecessary.
- **Activation visit at 4–6 weeks** once pain has resolved; **partner ideally attends** to learn pump operation.
- **Daily cycling** passively dilates cylinders and corpora and builds familiarity; **sexual use at 4–6 weeks.**

Mechanical Reliability: Repair for Mechanical Defect

- **Mechanical malfunction remains the most common complication** of penile prosthesis surgery.
- **Semirigid implants rarely fail mechanically**; long-term data are mostly for three-piece inflatables.
- **Failure modes: cylinder aneurysm, pump malfunction, autoinflation, tube kinking** — reduced by vendor modifications.
- **Long-term survival**: the **Mentor Alpha-1** (after its 1992 enhancement) and the **AMS 700CX/CXM** both showed **high freedom from mechanical failure at 5–10 years**.
- **Fluid loss is the commonest problem**, usually late; replace the leaking component if identified.
- **Implants in place more than 3 years → remove and replace the entire device** (wear on all parts).
- **If parts are left behind, confirm their hydraulic integrity** — a second defect may coexist: **hydraulic distension**, or more accurately an **ohmmeter test of electrical continuity**.

Ohmmeter Leak Test

Ohmmeter at most sensitive setting on nonsterile Mayo stand → sterilised cables passed off the field → 15-gauge blunt needle into the part's tubing; part filled with saline → one cable on the needle, the other on a retractor in the wound → leak → sodium and chloride ions conduct through the leak to the retractor → needle deflects; no leak → no deflection

- **Loss of rigidity with a semirigid implant → suspect fracture of the central metal core → explant and replace.**

Concept — Postoperative care shapes the **pseudocapsule**: inflated, upward-pointing, regularly cycled cylinders heal into a capsule that allows full expansion — the same capsule that causes length loss when neglected.

PATIENT AND PARTNER SATISFACTION

SATISFACTION BY DEVICE TYPE

Device	Series	Bottom line
AMS 700 three-piece	Several patient-partner series, including 1-year telephone follow-up	Most patients use the device regularly , rate erections excellent or satisfactory; most partners similarly satisfied ; small dissatisfied minority — insufficient rigidity and diminished length the key drivers
Ambicor two-piece	Two large series	High satisfaction; the later series slightly lower , partners lower than patients
Spectra rod; Spectra versus Genesis	Single series; comparative study	Favourable ; the two rods comparable

- **Pattern**: satisfaction is **high across all device types**, with **partners consistently slightly less satisfied than patients**.
- **Trap**: the drivers of dissatisfaction are **length and rigidity** — the same issues that dominate consent.
- **Complaints**: **perceived penile length, chronic pain or coolness, difficulty with mechanical operation, decreased sensation**.
- **Preoperative counselling is paramount**; some benefit from **sexual counselling before and after surgery**.
- **Advise that prostheses do not replicate normal erections** but provide **support and rigidity adequate for sexual activity**; **ejaculatory dysfunction and sensitivity changes** may occur.

Concept — Satisfaction is **decided at consent**: the dissatisfied minority are those whose expectations of length and natural feeling were not reset.

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